

GIRLS IN STEM

50 Awesome Science, Tech, Engineering
& Math Projects (Ages 8-13)

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GIRLS IN STEM

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BUILD A BRIDGE

Category: Engineering

Difficulty: Medium

Time: 30 min

Materials: Popsicle sticks, glue, coins for weight testing

STEPS:

1. Design your bridge on paper first
2. Build two towers using stacked popsicle sticks
3. Create a flat road surface connecting the towers
4. Add diagonal support beams for strength
5. Test by placing coins on the bridge until it breaks
6. Record how much weight it held

THE SCIENCE BEHIND IT:

Triangles are the strongest shape in engineering because they distribute force evenly. Real bridges use triangular trusses for this reason.

Careers: Civil Engineer, Structural Engineer, Architect

EGG DROP CHALLENGE

Category: Engineering

Difficulty: Hard

Time: 45 min

Materials: Raw egg, tape, paper, straws, cotton balls, plastic bag

STEPS:

1. Design a container to protect an egg from a 2-meter drop
2. Consider cushioning, air resistance, and shock absorption
3. Build your prototype using available materials
4. Test from a low height first
5. Modify your design based on results
6. Final test from full height

THE SCIENCE BEHIND IT:

Energy must be absorbed gradually to prevent damage. Crumple zones in cars work the same way - they deform slowly to protect passengers.

Careers: Aerospace Engineer, Product Designer, Safety Engineer

LEMON BATTERY

Category: Chemistry

Difficulty: Easy

Time: 20 min

Materials: Lemon, copper penny, zinc nail, LED light, wires

STEPS:

1. Roll the lemon to release juices inside
2. Insert the copper penny halfway into one side
3. Insert the zinc nail into the other side
4. Connect wires from each metal to the LED
5. Observe the LED light up dimly
6. Try multiple lemons in series for more power

THE SCIENCE BEHIND IT:

Chemical reactions between acid (lemon juice) and two different metals create an electrical current. This is how all batteries work - chemical energy converts to electrical energy.

Careers: Electrical Engineer, Chemist, Battery Researcher

INVISIBLE INK

Category: Chemistry

Difficulty: Easy

Time: 15 min

Materials: Lemon juice, cotton swab, white paper, heat source (lamp)

STEPS:

1. Dip cotton swab in lemon juice
2. Write a secret message on white paper
3. Let the paper dry completely
4. Hold paper near a warm lamp or iron on low
5. Watch the message appear in brown!
6. Try other liquids: milk, vinegar, honey

THE SCIENCE BEHIND IT:

Heat causes the carbon compounds in lemon juice to break down and oxidize, turning brown. This is the same chemical reaction that makes toast brown.

Careers: Forensic Scientist, Chemist, Spy Technologist

DENSITY TOWER

Category: Chemistry

Difficulty: Easy

Time: 20 min

Materials: Honey, corn syrup, dish soap, water, vegetable oil, rubbing alcohol, tall

STEPS:

1. Pour honey into the bottom of a tall clear glass
2. Slowly add corn syrup on top
3. Add dish soap gently
4. Pour water very slowly down the side
5. Add vegetable oil
6. Top with rubbing alcohol - observe layers!

THE SCIENCE BEHIND IT:

Density is how much mass fits in a given volume. Heavier liquids sink below lighter ones. Oil floats on water because it is less dense, which is why oil spills spread on ocean surfaces.

Careers: Chemical Engineer, Oceanographer, Materials Scientist

PAPER AIRPLANE SCIENCE

Category: Physics

Difficulty: Easy

Time: 30 min

Materials: Paper, tape measure, pencil, notebook

STEPS:

1. Fold 5 different paper airplane designs
2. Measure and mark a throwing line on the ground
3. Throw each design 3 times from the same spot
4. Measure and record each flight distance
5. Calculate the average for each design
6. Determine which design flies farthest and why

THE SCIENCE BEHIND IT:

Four forces act on a plane: thrust (your throw), lift (air under wings), gravity (pulling down), and drag (air resistance). Wing shape and weight distribution determine which force wins.

Careers: Aeronautical Engineer, Pilot, Aircraft Designer

CATAPULT BUILDER

Category: Engineering

Difficulty: Medium

Time: 25 min

Materials: Popsicle sticks, rubber bands, bottle cap, small objects to launch

STEPS:

1. Stack 5 popsicle sticks and band them together
2. Take 2 more sticks and band one end only
3. Wedge the stack between the two loose sticks
4. Glue a bottle cap to the top stick as the basket
5. Place a small object in the cap and press down
6. Release and measure how far the object flies

THE SCIENCE BEHIND IT:

Catapults store potential energy when you press down, then convert it to kinetic energy when released. The lever arm multiplies the force, launching objects farther than you could throw them.

Careers: Mechanical Engineer, Robotics Engineer, Physics Professor

CRYSTAL GROWING

Category: Chemistry

Difficulty: Easy

Time: 3 days

Materials: Sugar or borax, hot water, jar, string, pencil

STEPS:

1. Dissolve as much sugar or borax as possible in hot water
2. Tie a string to a pencil and suspend it in the solution
3. Place the jar somewhere undisturbed for 3 days
4. Check daily as crystals form on the string
5. Remove and examine crystal shapes with magnifying glass
6. Research why crystals form geometric shapes

THE SCIENCE BEHIND IT:

As water evaporates, dissolved molecules come out of solution and arrange themselves in repeating patterns. This orderly arrangement creates the geometric shapes of crystals. Every mineral forms unique crystal shapes.

Careers: Geologist, Materials Scientist, Mineralogist

BINARY CODE MESSAGES

Category: Coding

Difficulty: Medium

Time: 25 min

Materials: Paper, pencil, binary code chart

STEPS:

1. Learn that computers use only 1s and 0s (binary)
2. Write the alphabet in binary using 8-digit codes
3. Encode your name in binary (A=01000001)
4. Create a message for a friend to decode
5. Decode their message back to letters
6. Discuss how computers process billions of these per second

THE SCIENCE BEHIND IT:

All computer data (text, images, music, video) is stored as binary: sequences of 1s and 0s. Each 1 or 0 is called a bit. Eight bits make a byte, which can represent one character.

Careers: Software Developer, Data Scientist, Computer Engineer

ALGORITHM ART

Category: Coding

Difficulty: Easy

Time: 20 min

Materials: Graph paper, colored pencils, simple rules

STEPS:

1. Write 3 simple rules (e.g., color every 3rd square blue)
2. Add another rule (color squares in row 5 red)
3. Add a conditional rule (if blue AND row 5, make purple)
4. Follow your rules exactly to create a pattern
5. Try different rule combinations for new patterns
6. This is how computers follow code instructions!

THE SCIENCE BEHIND IT:

An algorithm is a set of step-by-step instructions. When you follow rules precisely without skipping steps, you are executing an algorithm - exactly what computers do with code.

Careers: Software Engineer, Game Developer, AI Researcher

SORTING RACE

Category: Coding

Difficulty: Easy

Time: 15 min

Materials: Deck of cards, timer

STEPS:

1. Shuffle a deck of cards completely
2. Time yourself sorting them in order using any method
3. Learn bubble sort: compare pairs, swap if needed, repeat
4. Learn selection sort: find the smallest, put it first, repeat
5. Time each method with a shuffled deck
6. Discuss which algorithm is faster and why

THE SCIENCE BEHIND IT:

Sorting algorithms are fundamental to computer science. Computers sort data billions of times daily for searches, databases, and recommendations. Efficiency matters when sorting millions of items.

Careers: Software Engineer, Database Administrator, Algorithm Designer

BALLOON ROCKET

Category: Physics

Difficulty: Easy

Time: 15 min

Materials: Balloon, string, straw, tape, two chairs

STEPS:

1. Tie string between two chairs across a room
2. Thread a straw onto the string
3. Blow up a balloon without tying it
4. Tape the inflated balloon to the straw
5. Release the balloon and watch it zoom along the string
6. Try different balloon sizes and measure speeds

THE SCIENCE BEHIND IT:

Newton's Third Law: for every action there is an equal and opposite reaction.

Air rushing out the back pushes the balloon forward. Real rockets work the same way with burning fuel.

Careers: Rocket Scientist, Aerospace Engineer, Propulsion Engineer

WATER FILTRATION

Category: Engineering

Difficulty: Medium

Time: 30 min

Materials: Plastic bottle, sand, gravel, cotton balls, dirty water, charcoal

STEPS:

1. Cut bottom off a plastic bottle
2. Layer cotton balls at the narrow end
3. Add a layer of fine sand above cotton
4. Add crushed charcoal above sand
5. Add gravel on top
6. Pour muddy water through and observe it come out clearer

THE SCIENCE BEHIND IT:

Filtration removes particles by trapping them in progressively finer materials. Charcoal also absorbs chemical contaminants. City water treatment plants use these same principles on a massive scale.

Careers: Environmental Engineer, Water Treatment Specialist, Hydrologist

MAGNETIC SLIME

Category: Chemistry

Difficulty: Medium

Time: 20 min

Materials: White glue, liquid starch, iron filings, magnet

STEPS:

1. Mix 1/4 cup white glue with 1/4 cup liquid starch
2. Knead until slime forms (add more starch if sticky)
3. Slowly add 2 tablespoons iron filings while kneading
4. Mix thoroughly until filings are distributed evenly
5. Hold a strong magnet near the slime
6. Watch the slime reach toward the magnet!

THE SCIENCE BEHIND IT:

Iron is a ferromagnetic material that responds to magnetic fields. When iron filings are suspended in slime, the whole mixture responds to magnets, stretching and moving toward the magnetic force.

Careers: Materials Scientist, Physicist, Nanotechnologist

SOLAR OVEN

Category: Engineering

Difficulty: Hard

Time: 45 min

Materials: Pizza box, aluminum foil, plastic wrap, black paper, tape

STEPS:

1. Cut a flap in the pizza box lid
2. Line the flap with aluminum foil (shiny side out)
3. Place black paper on the bottom of the box inside
4. Cover the opening with clear plastic wrap (creates greenhouse)
5. Prop the foil flap to reflect sun into the box
6. Place a s'more inside and wait for it to melt!

THE SCIENCE BEHIND IT:

The black surface absorbs solar radiation, the plastic wrap traps heat inside (greenhouse effect), and the foil flap reflects additional sunlight in. This is how solar energy works on a small scale.

Careers: Solar Engineer, Renewable Energy Researcher, Sustainability Engineer

DNA EXTRACTION

Category: Chemistry

Difficulty: Medium

Time: 25 min

Materials: Strawberry, dish soap, salt, rubbing alcohol, plastic bag, coffee filter

STEPS:

1. Place a strawberry in a plastic bag and mash it completely
2. Add a teaspoon each of dish soap and salt with some water
3. Mix gently for 2 minutes (soap breaks cell walls, salt clumps DNA)
4. Pour through coffee filter into a clear glass
5. Slowly pour cold rubbing alcohol on top
6. Watch white stringy DNA appear between the layers!

THE SCIENCE BEHIND IT:

Every living cell contains DNA. Soap dissolves the cell membrane, salt causes DNA strands to clump together, and alcohol causes DNA to precipitate out of solution so you can see it.

Careers: Geneticist, Molecular Biologist, Forensic Scientist

FIBONACCI IN NATURE

Category: Math

Difficulty: Easy

Time: 30 min

Materials: Notebook, pencil, flowers, pinecones, sunflowers (or photos)

STEPS:

1. Learn the Fibonacci sequence: 1, 1, 2, 3, 5, 8, 13, 21...
2. Count petals on different flowers (often 3, 5, 8, or 13)
3. Count spirals on a pinecone (usually 8 and 13)
4. Look at a sunflower center spirals (21 and 34)
5. Draw the Fibonacci spiral using squares
6. Find three more examples in nature

THE SCIENCE BEHIND IT:

The Fibonacci sequence appears everywhere in nature because it represents the most efficient growth pattern. Leaves arranged in Fibonacci spirals get maximum sunlight without shadowing each other.

Careers: Mathematician, Biologist, Data Pattern Analyst

PROBABILITY GAMES

Category: Math

Difficulty: Easy

Time: 20 min

Materials: Two dice, paper, pencil, 100 trials

STEPS:

1. Roll two dice and add the numbers together
2. Record the sum for 100 rolls
3. Create a bar graph of your results
4. Notice that 7 appears most often (predict why)
5. Calculate theoretical probability of each sum
6. Compare your experimental results to theory

THE SCIENCE BEHIND IT:

Probability theory predicts outcomes of random events. There are more ways to make 7 ($1+6$, $2+5$, $3+4$, $4+3$, $5+2$, $6+1$ = six ways) than any other sum. Casinos use probability to ensure they always win.

Careers: Statistician, Actuary, Data Scientist, Game Designer

TESSELLATION ART

Category: Math

Difficulty: Easy

Time: 25 min

Materials: Paper, scissors, tape, colored pencils

STEPS:

1. Cut a simple shape from one side of a square
2. Tape it to the opposite side (this creates a tessellation tile)
3. Trace your new shape repeatedly to fill a page
4. Make sure shapes fit together with no gaps or overlaps
5. Color your tessellation in a pattern
6. Research M.C. Escher's tessellation artwork

THE SCIENCE BEHIND IT:

Tessellations are patterns that cover a surface with no gaps or overlaps. Only certain shapes can tessellate (triangles, squares, hexagons). This math appears in tile floors, honeycomb, and Islamic art.

Careers: Mathematician, Graphic Designer, Architect, Tile Designer

CIPHER WHEEL

Category: Coding

Difficulty: Medium

Time: 20 min

Materials: Two cardboard circles (different sizes), brad fastener, marker

STEPS:

1. Cut two circles from cardboard (one smaller than the other)
2. Write alphabet around the edge of each circle
3. Attach centers with a brad so they can rotate
4. Choose a rotation key (shift by 3 letters = Caesar cipher)
5. Encode a message using your wheel
6. Give wheel and message to friend to decode

THE SCIENCE BEHIND IT:

Encryption converts readable text to code that only authorized people can read.

Caesar ciphers shift each letter by a fixed amount. Modern encryption uses much more complex math but the same principle.

Careers: Cybersecurity Analyst, Cryptographer, Information Security

HYDRAULIC ARM

Category: Engineering

Difficulty: Hard

Time: 45 min

Materials: Cardboard, syringes (no needles), plastic tubing, water, hot glue

STEPS:

1. Build a cardboard arm with joints that can bend
2. Connect syringes with plastic tubing and fill with water
3. Attach one syringe to each joint of the arm
4. When you push one syringe, water pushes the other
5. Practice picking up objects with your hydraulic arm
6. Add more syringes for more degrees of movement

THE SCIENCE BEHIND IT:

Hydraulics use incompressible fluid (water) to transmit force over distance.

When you push one syringe, equal force appears at the other. Excavators, car brakes, and elevators all use hydraulics.

Careers: Mechanical Engineer, Robotics Engineer, Heavy Equipment Designer

STATIC ELECTRICITY

Category: Physics

Difficulty: Easy

Time: 15 min

Materials: Balloon, wool cloth, small paper pieces, wall, water faucet

STEPS:

1. Rub balloon vigorously on wool for 30 seconds
2. Hold near small paper pieces and watch them jump up
3. Stick the balloon to a wall (it stays!)
4. Hold near a thin stream of water and watch it bend
5. Try rubbing different materials on the balloon
6. Research why rubbing creates static charge

THE SCIENCE BEHIND IT:

Rubbing transfers electrons from one surface to another, creating a negative charge on the balloon. This charge attracts neutral objects by inducing opposite charges on their surfaces.

Careers: Physicist, Electrical Engineer, Materials Scientist

PENDULUM WAVES

Category: Physics

Difficulty: Medium

Time: 30 min

Materials: String, nuts/washers, dowel rod, two supports

STEPS:

1. Cut strings at increasing lengths (20cm, 22cm, 24cm... up to 40cm)
2. Tie a washer to the end of each string
3. Attach all strings to the dowel rod, evenly spaced
4. Support the rod between two equal-height surfaces
5. Pull all washers back equally and release together
6. Watch the mesmerizing wave patterns that emerge

THE SCIENCE BEHIND IT:

Pendulum period depends on string length. Shorter pendulums swing faster. When many pendulums swing together with slightly different periods, they create beautiful wave patterns that cycle through chaos and alignment.

Careers: Physicist, Wave Researcher, Musician (sound waves)

PARACHUTE DESIGN

Category: Engineering

Difficulty: Medium

Time: 30 min

Materials: Plastic bags, string, tape, small toys, timer

STEPS:

1. Cut different-sized circles from plastic bags
2. Attach 4 equal strings to the edges of each circle
3. Tie strings to a small toy (equal weight for each test)
4. Drop from the same height and time each descent
5. Record which size parachute was slowest (most drag)
6. Try cutting a hole in the top and compare results

THE SCIENCE BEHIND IT:

Parachutes work by increasing air resistance (drag). The large surface area catches air, slowing descent. A hole in the top stabilizes the parachute by letting air pass through evenly.

Careers: Aerospace Engineer, Skydiving Equipment Designer, Military Engineer

CODING WITH CARDS

Category: Coding

Difficulty: Easy

Time: 20 min

Materials: Index cards, markers

STEPS:

1. Write simple commands on cards: Forward, Turn Left, Turn Right, Repeat
2. Create a course with obstacles (books, shoes)
3. Program a friend by giving them only your card instructions
4. They must follow EXACTLY what cards say (like a computer)
5. Debug when your friend goes wrong (find the error in your code)
6. Increase complexity with IF cards and WHILE loops

THE SCIENCE BEHIND IT:

Programming is writing precise instructions that a computer follows exactly.

Unlike humans, computers cannot interpret vague directions. This activity shows why precise, step-by-step thinking matters in coding.

Careers: Software Developer, QA Engineer, Robotics Programmer

EROSION EXPERIMENT

Category: Chemistry

Difficulty: Easy

Time: 25 min

Materials: 3 trays, soil, water, grass seeds, rocks, watering can

STEPS:

1. Fill three trays with soil at a slight angle
2. Leave tray 1 bare soil
3. Plant grass seeds in tray 2 (grow for a week first)
4. Add rocks and pebbles to tray 3
5. Pour water from the high end of each tray
6. Observe which tray loses the most soil to erosion

THE SCIENCE BEHIND IT:

Vegetation and rocks prevent erosion by slowing water flow and holding soil in place with roots. Deforestation causes massive erosion because tree roots no longer anchor the soil.

Careers: Environmental Scientist, Geologist, Conservation Engineer

FRACTAL DRAWING

Category: Math

Difficulty: Medium

Time: 30 min

Materials: Paper, ruler, pencil, colored pencils

STEPS:

1. Draw an equilateral triangle
2. Find the midpoint of each side and connect them (4 smaller triangles)
3. Remove the center triangle
4. Repeat for each remaining triangle
5. Continue for 4-5 iterations (Sierpinski Triangle)
6. Color each iteration a different shade

THE SCIENCE BEHIND IT:

Fractals are patterns that repeat at every scale - zoom in and you see the same shape. Nature is full of fractals: coastlines, snowflakes, broccoli, lightning, and blood vessels all show fractal patterns.

Careers: Mathematician, Computer Graphics Designer, Geographer

VOLCANO REACTION

Category: Chemistry

Difficulty: Easy

Time: 15 min

Materials: Baking soda, vinegar, dish soap, food coloring, container

STEPS:

1. Build a volcano shape from clay or a bottle
2. Put 2 tablespoons baking soda inside
3. Add a squirt of dish soap and food coloring
4. Pour vinegar in and step back
5. Observe the foaming eruption
6. Try different amounts and record which erupts most

THE SCIENCE BEHIND IT:

Baking soda (base) reacts with vinegar (acid) to produce carbon dioxide gas.

Dish soap traps the gas in bubbles, creating the foaming effect. This acid-base reaction is fundamental to chemistry.

Careers: Chemist, Volcanologist, Chemical Engineer

OPTICAL ILLUSIONS

Category: Math

Difficulty: Easy

Time: 20 min

Materials: Paper, ruler, pencil, colored pencils

STEPS:

1. Draw two horizontal lines of equal length with different arrow endings
2. Ask friends which looks longer (Mueller-Lyer illusion)
3. Draw a checkerboard pattern with two identical gray squares
4. Research why your brain misinterprets what your eyes see
5. Create your own optical illusion
6. Present to family and explain the science

THE SCIENCE BEHIND IT:

Optical illusions exploit shortcuts your brain uses to process visual information. Your brain makes assumptions about depth, color, and size that are usually correct but can be tricked.

Careers: Neuroscientist, Cognitive Psychologist, UX Designer

RUBBER BAND GUITAR

Category: Physics

Difficulty: Easy

Time: 20 min

Materials: Shoe box, rubber bands of different sizes, cardboard tube

STEPS:

1. Cut a hole in the lid of a shoe box
2. Stretch rubber bands of different thicknesses across the hole
3. Attach a cardboard tube as the neck
4. Pluck each rubber band and listen to different pitches
5. Observe: thin/tight bands = high pitch, thick/loose = low
6. Try changing tension and length to tune your guitar

THE SCIENCE BEHIND IT:

Sound is vibration. Thinner, tighter, or shorter strings vibrate faster (higher frequency = higher pitch). This is the physics behind every stringed instrument from guitars to pianos.

Careers: Acoustic Engineer, Music Producer, Sound Designer

ELECTROMAGNET

Category: Physics

Difficulty: Medium

Time: 20 min

Materials: Iron nail, insulated wire, D battery, paper clips

STEPS:

1. Wrap insulated wire around an iron nail 50+ times
2. Strip the wire ends to expose copper
3. Connect both wire ends to a D battery
4. Hold the nail near paper clips - they stick!
5. Try different numbers of wire wraps
6. Disconnect battery when done (wire gets hot)

THE SCIENCE BEHIND IT:

Electric current flowing through wire creates a magnetic field. Wrapping wire around iron concentrates this field, making a magnet you can turn on and off. MRI machines use powerful electromagnets.

Careers: Electrical Engineer, MRI Technician, Physicist

PIXEL ART MATH

Category: Math

Difficulty: Easy

Time: 25 min

Materials: Graph paper, colored pencils, coordinate system

STEPS:

1. Draw x and y axes on graph paper
2. Plot points at coordinates to create a picture
3. Color in squares at each coordinate
4. Create instructions (coordinates) for a friend to follow
5. See if their picture matches yours
6. Research how computer screens display images as pixels

THE SCIENCE BEHIND IT:

Computer screens display images as grids of tiny colored squares (pixels). Each pixel has a coordinate and a color value. Understanding coordinate systems is essential to computer graphics.

Careers: Game Developer, Digital Artist, Computer Graphics Engineer

ECOSYSTEM IN A JAR

Category: Chemistry

Difficulty: Easy

Time: 30 min + weeks of observation

Materials: Large glass jar, soil, small plants, water, insects (optional), plastic

STEPS:

1. Layer small rocks then soil in a large glass jar
2. Plant 2-3 small plants
3. Add a small amount of water
4. Seal with plastic wrap (creates closed ecosystem)
5. Place in indirect sunlight
6. Observe for weeks - plants cycle water and air themselves

THE SCIENCE BEHIND IT:

In a sealed ecosystem, plants produce oxygen and absorb CO₂, while decomposers break down dead material into nutrients. Water cycles through evaporation and condensation. This mimics Earth's systems.

Careers: Ecologist, Environmental Scientist, Biosphere Researcher

LOGIC GATE PUZZLES

Category: Coding

Difficulty: Medium

Time: 20 min

Materials: Paper, pencil, flashlights (optional)

STEPS:

1. Learn AND gate: both inputs must be ON for output ON
2. Learn OR gate: either input ON = output ON
3. Learn NOT gate: reverses the input
4. Combine gates to solve logic puzzles
5. Use flashlights to physically demonstrate each gate
6. Build a truth table for each combination

THE SCIENCE BEHIND IT:

Logic gates are the building blocks of all computer chips. Every calculation your computer makes is done using combinations of these simple gates. A modern chip has billions of them.

Careers: Computer Engineer, Chip Designer, Digital Electronics Engineer

RUBE GOLDBERG MACHINE

Category: Engineering

Difficulty: Hard

Time: 60 min

Materials: Dominoes, marbles, cardboard tubes, ramps, cups, string

STEPS:

1. Plan a chain reaction with at least 8 steps
2. Start simple: domino knocks ball, ball rolls down ramp
3. Add more steps: ball hits cup, cup pulls string, etc.
4. Build each step and test individually
5. Connect all steps and test the full chain
6. Modify and troubleshoot until the whole machine works

THE SCIENCE BEHIND IT:

Rube Goldberg machines demonstrate energy transfer and chain reactions. Each step converts one type of energy to another. Engineers use chain-reaction thinking to design complex automated systems.

Careers: Mechanical Engineer, Automation Engineer, Systems Designer

SECRET CODE APP DESIGN

Category: Coding

Difficulty: Medium

Time: 30 min

Materials: Paper, pencil, colored pencils

STEPS:

1. Design a simple app on paper (wireframe) for sending secret messages
2. Draw each screen the user would see
3. Map the flow: how does the user move between screens?
4. Design the encoding algorithm (how messages are scrambled)
5. Add a decoding feature for received messages
6. Consider: what makes an app easy to use?

THE SCIENCE BEHIND IT:

App design involves User Interface (what it looks like) and User Experience (how it works). Designers sketch wireframes before any code is written. This planning prevents expensive mistakes later.

Careers: UX Designer, App Developer, Product Manager

PLANT GROWTH EXPERIMENT

Category: Chemistry

Difficulty: Easy

Time: 2 weeks

Materials: 6 small pots, bean seeds, water, measuring cup, ruler, notebook

STEPS:

1. Plant 2 seeds each in 6 identical pots with same soil
2. Give pots different amounts of water: none, little, medium, lots
3. Keep all other conditions the same (same light, temperature)
4. Measure and record plant height daily for 2 weeks
5. Create a graph of growth vs water amount
6. Determine the optimal watering amount

THE SCIENCE BEHIND IT:

This is the scientific method in action: one variable (water) changes while all others stay constant. This controlled experiment reveals cause and effect - the foundation of all science.

Careers: Botanist, Agricultural Scientist, Research Scientist

PAPER CIRCUIT CARDS

Category: Engineering

Difficulty: Medium

Time: 25 min

Materials: Copper tape, coin battery, LED light, cardstock, tape

STEPS:

1. Design a greeting card that lights up
2. Plan where the LED and battery will go
3. Lay copper tape in a circuit from battery to LED and back
4. Make sure tape has no gaps (breaks the circuit)
5. Fold the card so closing it presses battery against tape
6. Decorate your illuminated card!

THE SCIENCE BEHIND IT:

Electrical circuits need a complete loop from battery positive to negative. Any gap breaks the circuit and stops current flow. Switches work by intentionally creating or closing gaps.

Careers: Electrical Engineer, Circuit Designer, Electronics Maker

REACTION TIME TEST

Category: Physics

Difficulty: Easy

Time: 15 min

Materials: Ruler, friend, notebook, calculator

STEPS:

1. Have a friend hold a ruler vertically at the 0 end
2. Position your fingers at the 30cm mark without touching
3. Friend drops ruler without warning - catch it as fast as possible
4. Record where you caught it (distance = reaction time indicator)
5. Repeat 10 times and calculate average
6. Test: does practice improve reaction time?

THE SCIENCE BEHIND IT:

Your brain must detect the ruler dropping (eyes), send signal to brain, process it, send signal to muscles, then muscles contract. This entire neural pathway takes 0.15-0.3 seconds.

Careers: Neuroscientist, Sports Scientist, Human Factors Engineer

STRAW ROCKET

Category: Engineering

Difficulty: Easy

Time: 20 min

Materials: Paper, straws of 2 sizes, tape, scissors, ruler

STEPS:

1. Roll paper tightly around the larger straw to make a tube
2. Tape it shut and fold/tape one end closed (the rocket)
3. Add paper fins for stability
4. Slide rocket onto the smaller straw
5. Blow hard through the small straw to launch
6. Modify fin size and angle to improve distance and accuracy

THE SCIENCE BEHIND IT:

Fins stabilize rockets by keeping the center of drag behind the center of mass, preventing tumbling. Real rockets use the same principle. The air pressure from blowing provides thrust.

Careers: Rocket Engineer, Aerospace Designer, Wind Tunnel Researcher

GEOMETRIC ORIGAMI

Category: Math

Difficulty: Medium

Time: 30 min

Materials: Square paper, instructions for modular origami

STEPS:

1. Fold a single unit from a square of paper (Sonobe unit)
2. Make 6 identical units following exact measurements
3. Assemble units into a cube by interlocking tabs and pockets
4. Observe the geometry: each face is a square, angles are 90 degrees
5. Try 12 units for a more complex shape (stellated octahedron)
6. Research what Platonic solids are and which ones you can make

THE SCIENCE BEHIND IT:

Modular origami demonstrates geometric principles: symmetry, angles, faces, edges, and vertices. Mathematicians use paper folding to prove theorems and solve problems that seem unrelated to paper.

Careers: Mathematician, Structural Engineer, Architect, Industrial Designer

WEBSITE WIREFRAME

Category: Coding

Difficulty: Easy

Time: 25 min

Materials: Large paper, pencil, ruler, colored pencils

STEPS:

1. Choose a topic for a simple website (your hobby, a cause you care about)
2. Sketch the homepage layout: header, navigation, content, footer
3. Design 3 additional pages connected by navigation links
4. Add buttons, images (draw placeholders), and text areas
5. Think about: what does the user want to find? Make it easy!
6. Label which sections use HTML, which need CSS styling

THE SCIENCE BEHIND IT:

All websites start as wireframes - simple sketches showing layout and structure before any code is written. This planning ensures good user experience and saves hours of recoding later.

Careers: Web Developer, UX Designer, Front-End Engineer

YOU ARE A SCIENTIST!

Keep experimenting, building, and questioning.

The world needs YOUR innovations.

PERSONAL GROWTH JOURNAL

What resonated with me most:

Actions I will take:

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REFLECTION & ACTION STEPS

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Use this space for your thoughts and reflections:

GROWTH JOURNAL

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MY ACTION PLAN

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DEEPER EXPLORATION

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MY ACTION PLAN

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